Predicting Severe COVID-19 Infection Using Logistic Regression, Boosting, and Support Vector Machines

Introduction

This study aimed to identify important demographic and clinical predictors of severe COVID-19 infection and compare the predictive performance of advanced machine learning approaches, including boosting and support vector machines (SVM), with logistic regression. The analysis also evaluated whether predicted probabilities could be used as individualized risk scores for severe infection. A random sample of 1,000 participants was selected from the full dataset using a fixed random seed. The dataset included demographic characteristics and clinical measurements collected from participants infected with COVID-19 between 2021 and 2023. The outcome variable was COVID-19 severity, coded as severe versus not severe infection.

Exploratory Analysis

The dataset was checked for no missing value, then categorical variables were converted into factor variables with descriptive labels before analysis. This made the plots and model outputs easier to interpret and ensured that categorical predictors were properly treated in the models.

The dataset was randomly divided into training (80%) and testing (20%) sets using stratified sampling to preserve the proportion of severe infections. Exploratory analyses included summary statistics, a correlation matrix for continuous variables, boxplots for continuous predictors, and bar plots for categorical predictors.

The correlation matrix of continuous variables is shown in Figure 1. Weight and BMI were strongly positively correlated, while height and BMI were moderately negatively correlated. Since BMI is directly calculated from height and weight, including all three variables in the same model would introduce redundancy and potential multicollinearity. Therefore, height and weight were excluded from subsequent analyses while BMI was kept as the primary body measure. A moderate positive correlation was also observed between age and systolic blood pressure (SBP). Boxplots comparing continuous predictors by COVID-19 severity are shown in Figure 2. Participants with severe infection generally had higher age, BMI, SBP, and slightly higher LDL

cholesterol levels compared with participants without severe infection. Depression scores appeared relatively similar between the two groups.

Figure 3 presents combined bar plots for categorical predictors. Current smokers and unvaccinated participants showed noticeably higher proportions of severe infection compared with never smokers and vaccinated participants. Participants with hypertension also appeared to have higher rates of severe infection, while gender, race, and diabetes showed weaker visual relationships with severe infection.

Logistic Regression Model

A multivariable logistic regression model was fit using demographic and clinical predictors. Vaccination status was the strongest protective factor against severe infection, with vaccinated participants having substantially lower odds of severe COVID-19 infection compared with unvaccinated participants (OR = 0.029).

In contrast, current smoking status showed the largest increase in risk, with current smokers having approximately three times the odds of severe infection compared with individuals who never smoked (OR = 3.05). Higher BMI (OR = 1.30), older age (OR = 1.09), and higher systolic blood pressure (OR = 1.10) were also associated with increased odds of severe infection. Figure 4 further shows that smoking had the largest odds ratio greater than 1, while vaccination had an odds ratio far below 1, indicating a strong protective effect.

Variables such as gender, race, hypertension, diabetes, LDL cholesterol, and depression score were not statistically significant after adjustment for other predictors. The ROC curve for logistic regression is shown in Figure 5. The model achieved an AUC of 0.885, indicating strong predictive performance. The confusion matrix in Figure 6 shows that the model correctly classified most observations, including 124 non-severe cases and 41 severe cases. Overall, logistic regression achieved an accuracy of 0.821, sensitivity of 0.641, and specificity of 0.905.

Boosting Model

A gradient boosting machine model was trained using 5-fold cross-validation with tuning performed over multiple combinations of boosting iterations, tree depth, and shrinkage values. Cross-validation tuning results are shown in Figure 7. Performance generally improved during the early boosting iterations before plateauing or slightly decreasing at larger numbers of trees, suggesting potential overfitting with excessive boosting iterations. The optimal model used 500 trees, interaction depth of 4, and shrinkage of 0.005.

The ROC curve for the boosting model is shown in Figure 8. The model achieved an AUC of 0.890, slightly outperforming logistic regression. The confusion matrix shown in Figure 9 demonstrates that the boosting model achieved high specificity while correctly identifying a moderate proportion of severe infections. The boosting model achieved an overall accuracy of 0.841, sensitivity of 0.656, and specificity of 0.927.

Variable importance results are displayed in Figure 10. Vaccination status was by far the most influential predictor, with a relative influence of approximately 55.1. BMI and SBP were the next most important predictors, with relative influence values of about 16.0 and 15.7, respectively, followed by age at 6.8. Variables such as race, hypertension, and diabetes had very small relative influence values close to 0, indicating relatively limited contribution to the boosting model.

Support Vector Machine Model

A radial kernel support vector machine (SVM) model was trained using cross-validation with tuning over multiple gamma and cost values. Figure 11 displays the cross-validation tuning results for the SVM model. Lower error regions were concentrated around moderate gamma values and relatively small cost values, indicating improved classification performance within this tuning range. The optimal model was selected using gamma = 0.0498 and cost = 1.

The ROC curve for the SVM model is shown in Figure 12. The SVM model achieved the strongest predictive performance among all approaches, with an AUC of 0.894. The confusion matrix in Figure 13 demonstrates strong classification performance with relatively balanced prediction of severe and non-severe infections. The SVM model achieved an overall accuracy of 0.851, sensitivity of 0.703, and specificity of 0.920.

Permutation-based variable importance results for SVM are displayed in Figure 14. Similar to the boosting model, vaccination status was overwhelmingly the most important predictor, followed by SBP, BMI, and age. The consistency of important predictors across all models strengthens confidence in these findings.

Conclusion

All three models demonstrated strong predictive performance for severe COVID-19 infection (Table 1). The machine learning methods slightly outperformed logistic regression, although the differences were relatively small. Logistic regression achieved an AUC of 0.885, while boosting achieved the highest specificity (0.927). SVM achieved the strongest overall performance, with the highest accuracy (0.851), highest sensitivity (0.703), and highest AUC (0.894). These findings suggest that, in this dataset, logistic regression captured much of the predictive signal while remaining more interpretable than the machine learning approaches.

The predicted probabilities generated from the logistic regression, boosting, and SVM models can serve as individualized risk scores for estimating a participant's probability of severe COVID-19 infection based on demographic and clinical characteristics.

Across all models, vaccination status was consistently the strongest predictor of severe infection. Higher BMI, elevated SBP, older age, and current smoking status were also associated with increased risk, while vaccination substantially reduced the risk of severe infection.

Limitations

This study has several limitations. First, the analysis used a random sample of 1,000 participants rather than the full dataset, so results may vary slightly with different samples. Second, the data were observational, meaning the identified associations should not be interpreted as causal relationships. Third, external validation was not performed, so model performance may differ when applied to independent populations.

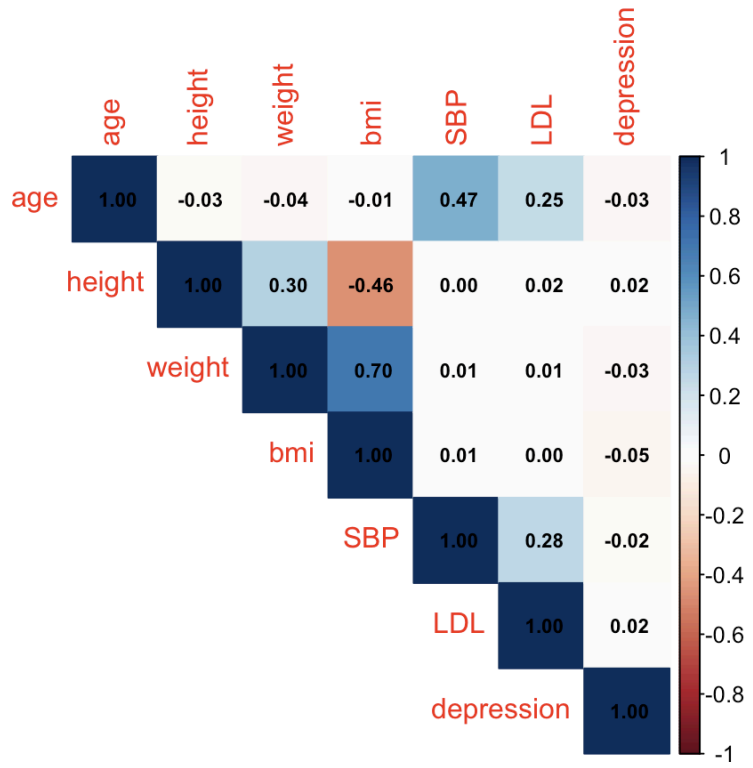


Figure 1: Correlation matrix of continuous predictors

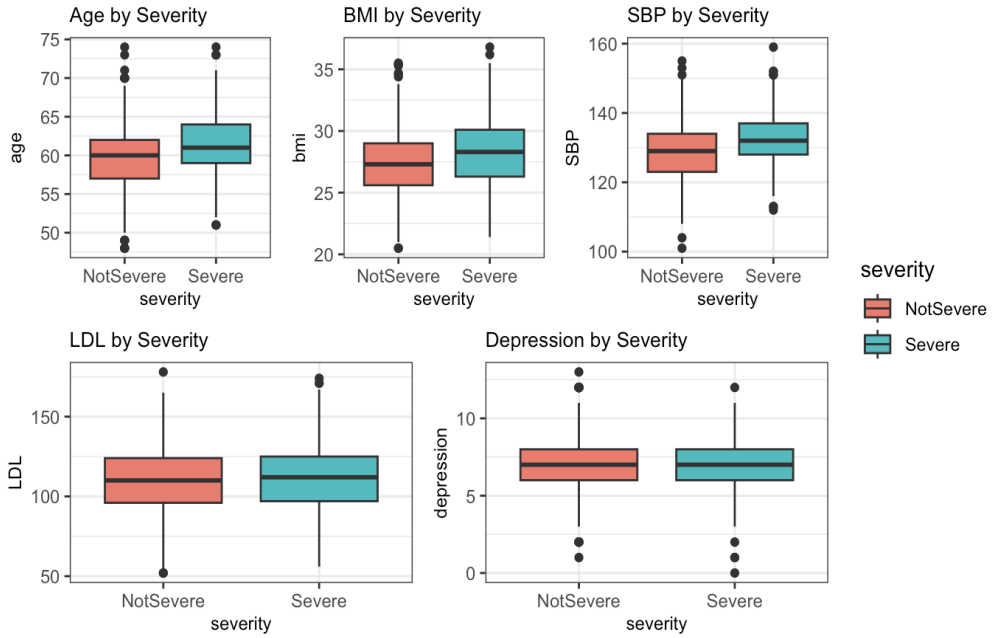


Figure 2: Boxplots of continuous predictors by COVID-19 severity

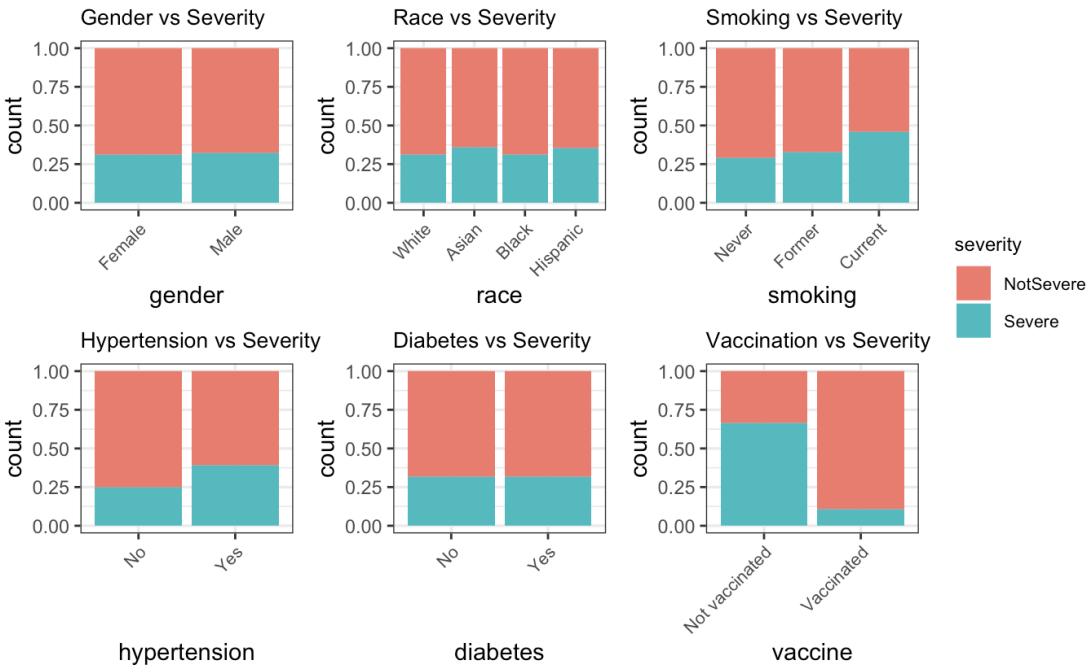


Figure 3: Categorical predictors by COVID-19 severity

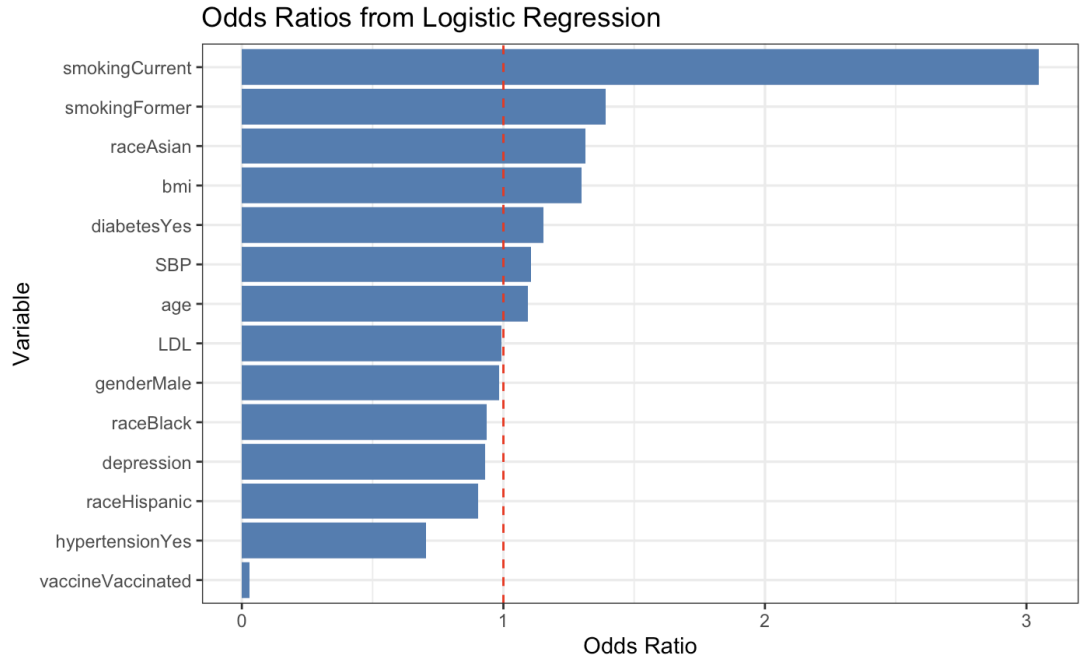


Figure 4: Odds ratios from logistic regression

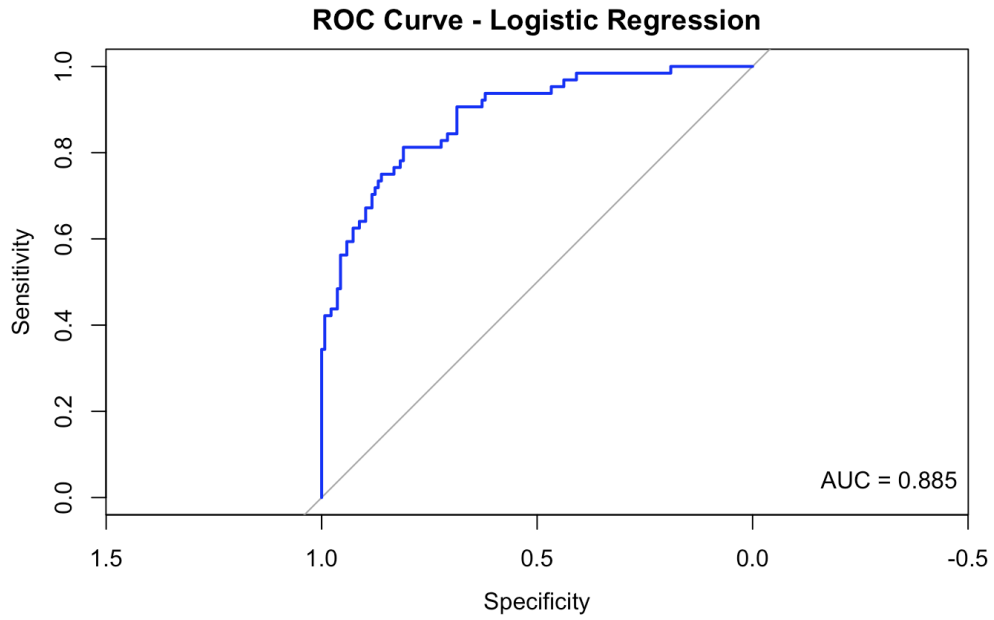


Figure 5: ROC curve for logistic regression

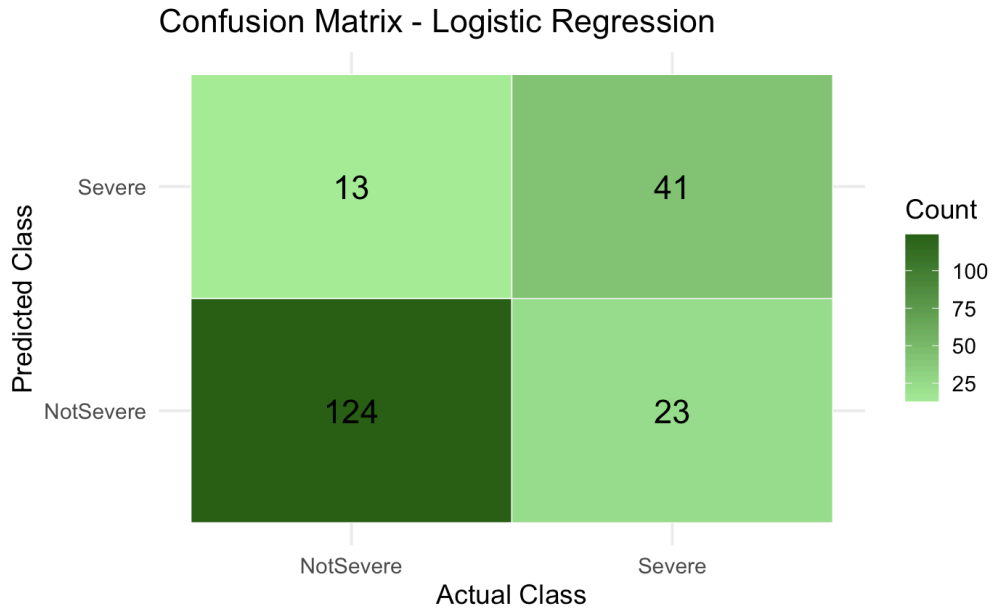


Figure 6: Confusion matrix for logistic regression

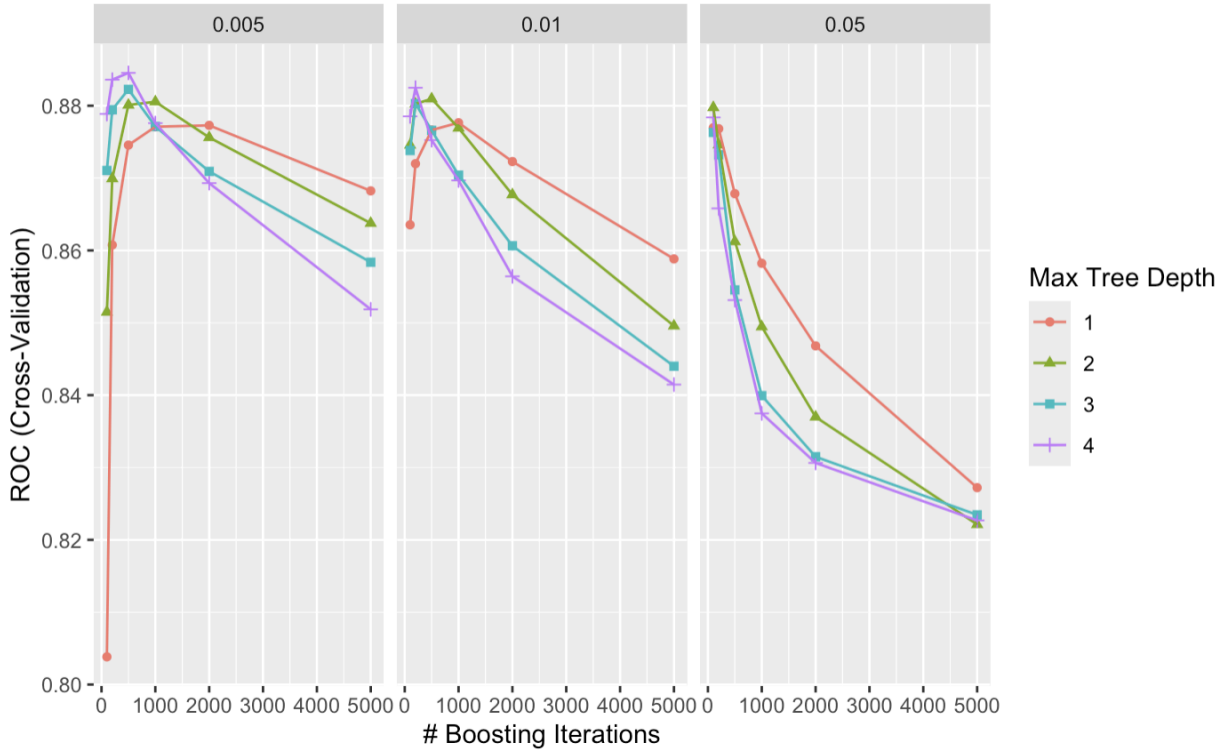


Figure 7: Cross-validation results for boosting

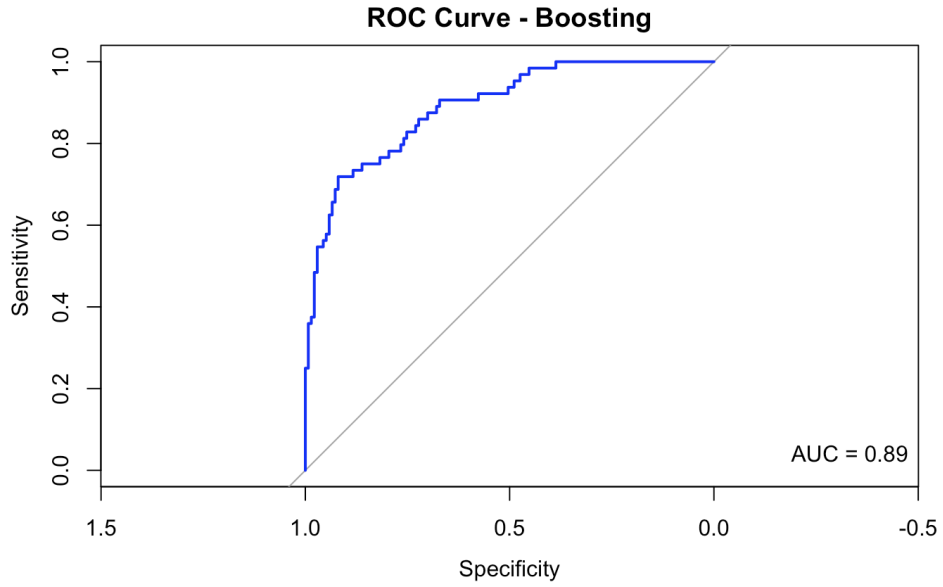


Figure 8: ROC curve for boosting

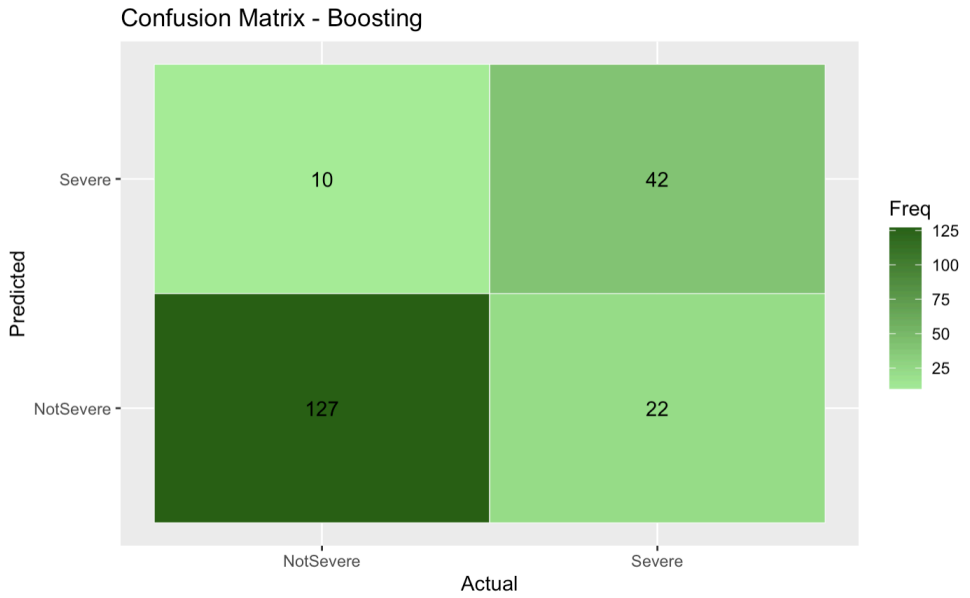


Figure 9: Confusion matrix for boosting

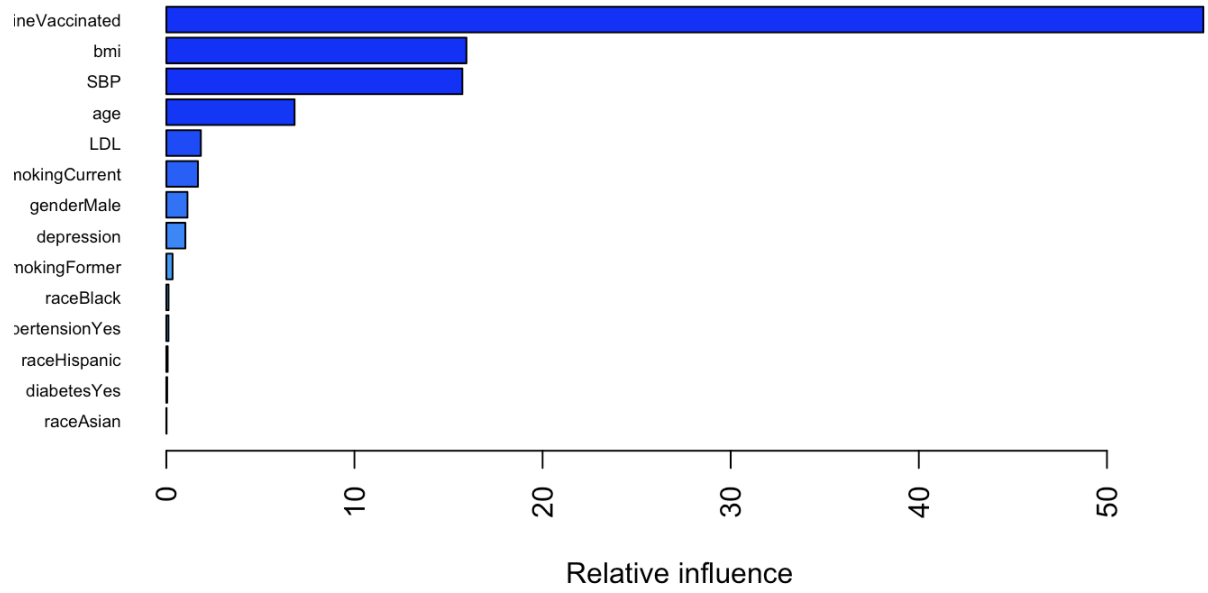


Figure 10: Variable importance for boosting

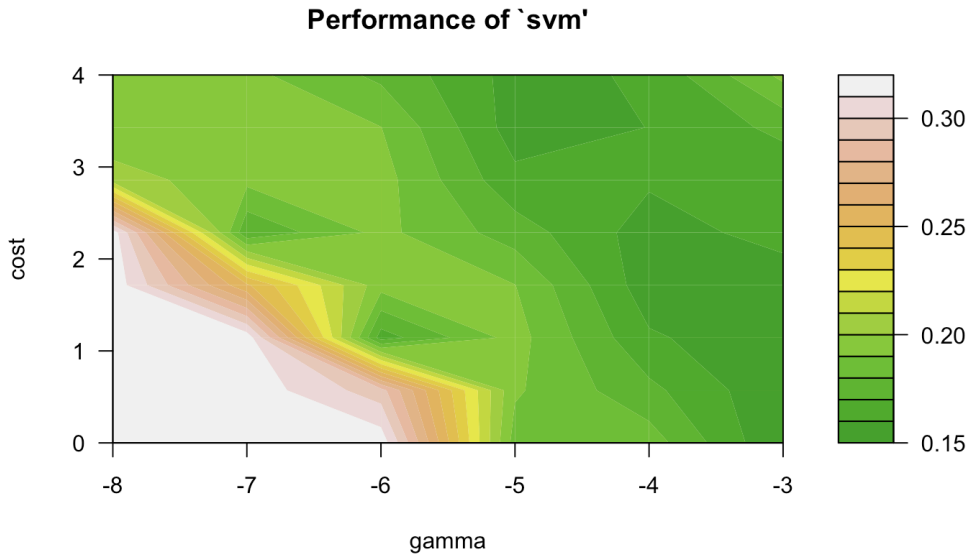


Figure 11: Cross-validation results for SVM

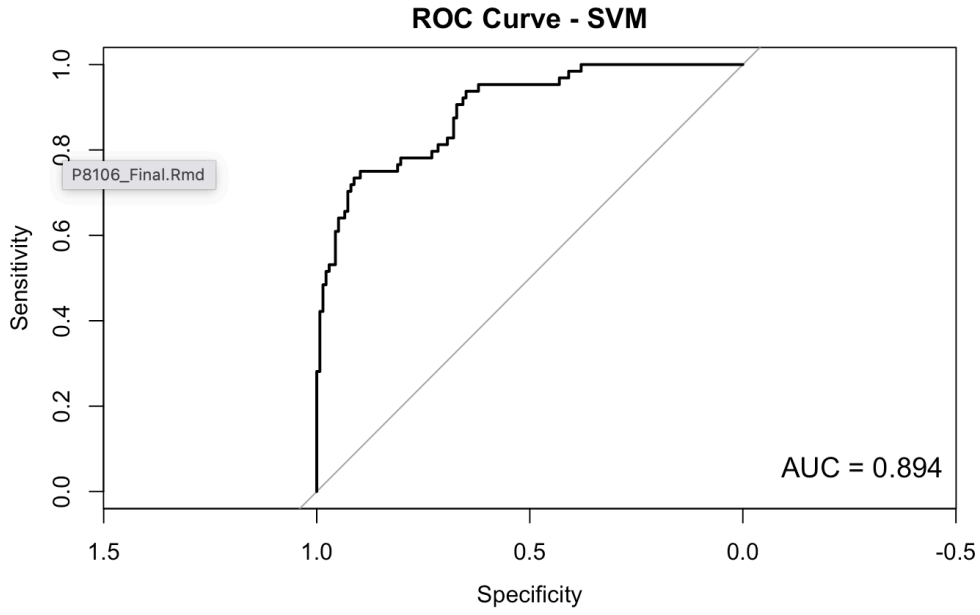


Figure 12: ROC curve for SVM

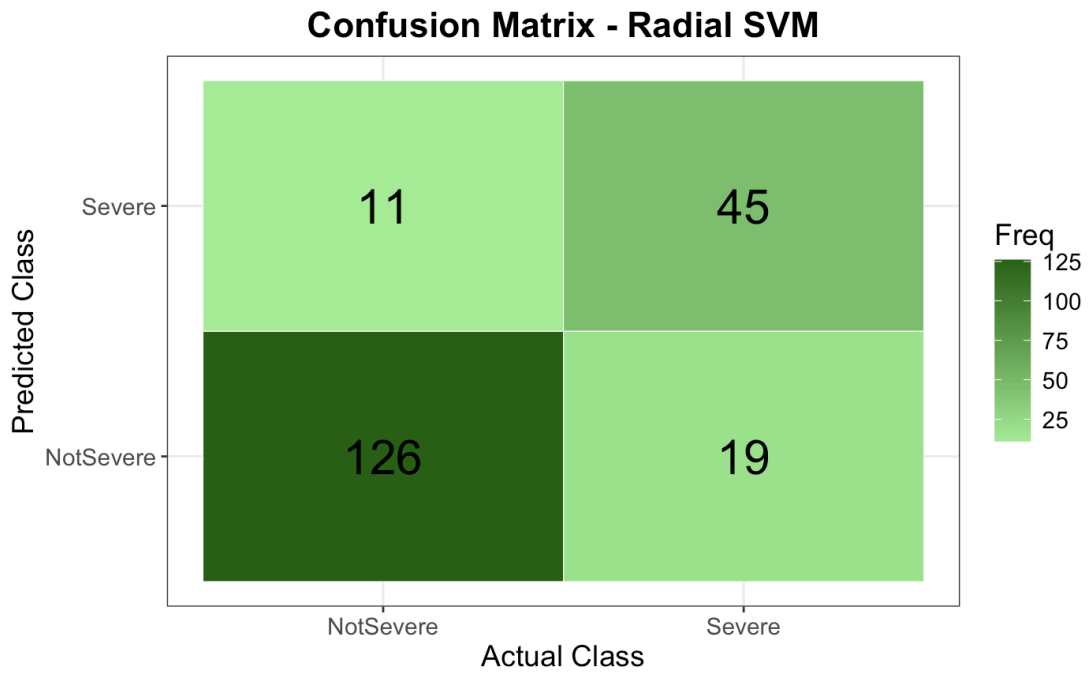


Figure 13: Confusion matrix for SVM

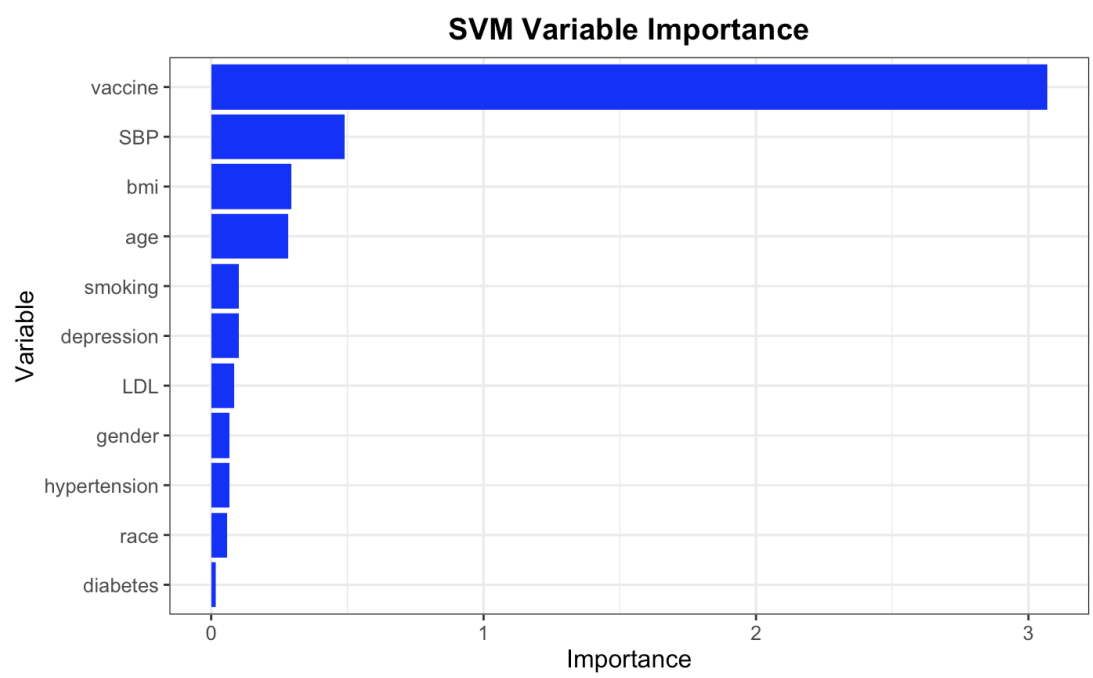


Figure 14: Variable importance for SVM

Table 1: Model Comparison

Model	Accuracy	Sensitivity	Specificity	AUC
Logistic Regression	0.821	0.641	0.905	0.885
Boosting	0.841	0.656	0.927	0.890
SVM	0.851	0.703	0.920	0.894

Table 1: Model performance comparison